



ASHAIN SIRIWARDANE

Associate Tech Lead

+94-77-2005661 | ashain.siriwardana@gmail.com | linkedin.com/in/ashain-siriwardane-1a861785 | github.com/sinfulexiled | sinfulexiled.github.io/MyPortfolio2025

PROFESSIONAL SUMMARY

Associate Tech Lead with 6+ years of experience designing and delivering scalable SaaS, products, e-commerce, enterprise integrations, and cloud-native systems. Proven track record of leading engineering teams of 3–5 developers, driving system architecture decisions, and mentoring junior engineers. Deep expertise in React.js, Node.js, TypeScript, and cloud platforms (AWS, Azure). Recently expanded into AI-powered systems including Retrieval-Augmented Generation (RAG) pipelines. Passionate about building clean, maintainable systems and translating complex business requirements into robust technical solutions.

CORE COMPETENCIES

Front-End: React.js, Next.js, Redux, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Material UI, Ant Design

Back-End: Node.js, Express.js, NestJS, REST APIs, GraphQL, Spring Boot, Workflow Automation, Systems Integration

AI & LLM: Large Language Models (LLMs), GPT, Claude, Retrieval-Augmented Generation (RAG), Vector Embeddings, Similarity Search, AI Pipeline Design, Google Genkit

Databases: PostgreSQL, MySQL, MongoDB, Firebase, Redis, Vector Databases

Cloud & DevOps: AWS (S3, EC2, Lambda), Azure (Service Bus, Durable Functions, Monitor), Google Cloud, Docker, CI/CD

Architecture: Micro-Frontends, Microservices, Multi-Tenant SaaS, Mono-Repo, Event-Driven, Orchestration, Low-Code, Single Responsibility Principle

Leadership & Process: Team Leadership, Code Review, Mentoring, Agile/Scrum, TDD, Stakeholder Management, System Design

Tools: Git, GitHub, Bitbucket, Jira, VS Code, Stripe, Socket.io

WORK EXPERIENCE

Allion Technologies

2025 – Present

Associate Tech Lead – Insurance Finance / Banking Integration

- Architected and built a low-code, config-driven orchestration platform from scratch that automates insurance claims flows, banking transactions, and data pipelines across enterprise systems — eliminating the need for individual point-to-point integrations.
- Engineered the orchestration core using Azure Service Bus (ASB) and Azure Durable Functions, enabling reliable, stateful workflow execution with built-in replay capability so transactions can be re-triggered from any upstream source without data loss.
- Platform has successfully orchestrated thousands of transactions across distributed financial systems, with workflows defined entirely through JSON/YAML configuration — removing the need for custom code per integration.

- Designed and documented an end-to-end AI pipeline leveraging Large Language Models (LLMs — specifically GPT and Claude) combined with Retrieval-Augmented Generation (RAG) techniques, vector embeddings, and similarity search to deliver intelligent, citation-backed responses from proprietary organisational data.
- Designed and delivered React.js front-end tooling and UI/UX for the platform, enabling non-engineering stakeholders to configure and monitor workflows without writing code.
- Led client requirement gathering and translated complex financial domain logic into orchestration rules, driving execution from architecture through production delivery.

Full-Stack Engineer – Wagyu Premium Food E-Commerce Platform

- Architected and delivered a premium Wagyu beef food e-commerce platform with multi-role support (Admin, Supplier, Customer) and a multi-stage order workflow spanning browsing, cart, payment, and fulfilment through to delivery.
- Built the full-stack system using Next.js 15 with Turbopack (frontend), NestJS (backend), PostgreSQL, and Redis caching, with Stripe payment processing and webhook validation for reliable transaction handling.
- Implemented a supplier product submission and approval workflow with pricing history tracking, stock management, and admin-controlled markup configuration for the marketplace model.
- Integrated real-time notifications via Socket.io, AI-powered product search and recommendations using Google Genkit, and next-intl internationalisation for multilingual market support.
- Designed a monorepo architecture using pnpm workspaces with shared TypeScript types and DTOs across frontend, backend, and database packages, ensuring type safety end-to-end.

Softlogic / PelicanCube

2025

Free Lance Consultant – Sri Lanka Railways Digital Platform

- Developed the Sri Lanka Railways ticketing, seat reservation, back-office management, and customer portal system using Next.js (front-end) and NestJS (back-end).
- Built the seat reservation engine handling real-time availability, booking flows, and ticket management for train routes across the national railway network.
- Designed and implemented the customer-facing portal enabling passengers to search routes, reserve seats, and manage bookings online.
- Developed the back-office management system for railway staff and administrators to manage schedules, inventory, and reservations.

Velaris

2022 – 2025

Senior Software Engineer – SaaS Customer Success Platform

- Owned and delivered multiple core product areas end-to-end across UI, API, and service layers on a B2B SaaS platform — including Success Plans, Task Management, Milestones, Files, Notes, and a drag-and-drop Scenario Workflow Builder.
- Built the drag-and-drop Scenario feature enabling customer success managers to visually construct automated workflows — a flagship product capability — using React.js with full backend integration via Node.js and PostgreSQL on AWS.
- Took full-stack ownership of the Task Handling system for organisational managers, designing the data model, REST API, service logic, and React UI from scratch.
- Architected and implemented the Success Plans and Milestones modules, providing customers with structured frameworks to track and manage client outcomes within the platform.
- Collaborated within a cross-functional team of 7–10 engineers, contributing to system design decisions, API contracts, and front-end architecture standards across the product.
- Mentored and coached junior and mid-level developers through code reviews and pair programming, raising overall code quality and engineering standards across the team.

- Managed PostgreSQL schemas and AWS infrastructure for owned product areas, ensuring performant queries and reliable cloud deployments.

OctopusBI

2021 – 2022

Software Engineer – SaaS Web Applications & Educational Dashboards

- Built and maintained analytics dashboards adopted by 200+ schools across Australia, serving 2,000+ users including teachers, administrators, and school managers tracking institutional performance metrics.
- Delivered per-school dashboard customisation capabilities, enabling each institution to configure views and metrics — integrated with Canvas LMS, Google Classroom, and Volkswagen's internal learning platform.
- Developed complex React.js applications using Redux Thunk and ducks architecture for predictable, scalable state management across a large multi-tenant front-end codebase.
- Implemented Next.js server-side rendering to improve initial page load performance and SEO across the platform.
- Designed and implemented app-wide custom authentication and role-based access control (RBAC), securing sensitive student and institutional data across all tenants.

SIUEX (Personal Venture)

2020 – 2021

Founder & Software Engineer – Enterprise Management Systems

- Delivered end-to-end design and development of Hospital, Event, and Restaurant Management Systems using microservices architecture and headless CMS.
- Built a Hospital Management System backend in Spring Boot, implementing IIT health data exchange standards for interoperability with external health platforms.
- Integrated Google Maps API for real-time navigation and location-based features within the event management product.

SiuexMedical – Medical SaaS Platform (Optical & Pharmacy)

- Designed and developed a full-stack medical SaaS platform for optical stores and pharmacies, supporting seamless dual-mode switching between optical and pharmacy workflows with mode-specific UIs, products, and data isolation per organisation.
- Built the frontend with React 18, Vite, Zustand, and Tailwind CSS with Radix UI, delivering prescription management, order lifecycle tracking, patient records, and a Canvas-based eye annotation tool for documenting patient examinations.
- Engineered a multi-tenant Node.js/Express/PostgreSQL backend with mode-aware repository pattern, isolating optical and pharmacy data per organisation to prevent cross-tenant data leakage.
- Integrated Anthropic Claude for AI-powered medical insights and recommendations, Stripe for subscription billing and plan management, and Nodemailer for automated patient and order communications.
- Implemented a full inventory management system for frames, lenses, and medications with real-time stock tracking, order-driven stock decrement, and configurable product catalogues per mode.

KEY PROJECTS

Low-Code Integration Orchestration Platform (Allion Technologies): Config-driven (JSON/YAML) orchestration engine using Azure Service Bus and Azure Durable Functions, automating insurance claims, banking transactions, and cross-system data pipelines with replay support. Orchestrated thousands of transactions. Stack: Node.js, Azure, React.js.

LLM / AI Pipeline for Organisational Knowledge (Allion Technologies): Architected an AI-powered natural language search system using LLMs (Claude & GPT) and Retrieval-Augmented Generation (RAG),

with vector embeddings and similarity search, enabling citation-backed responses from proprietary insurance and banking data. Stack: Node.js, Azure, Vector DB, Anthropic APIs.

Wagyu Premium Food E-Commerce Platform (Allion Technologies): Full-stack multi-role food e-commerce marketplace for premium Wagyu beef with supplier approval workflows, Stripe payments, real-time Socket.io notifications, and AI-powered search via Google Genkit. Stack: Next.js 15, NestJS, PostgreSQL, Redis, Stripe, Monorepo (pnpm workspaces).

SiuxMedical – Medical SaaS Platform (SIUEx): Dual-mode medical management platform for optical stores and pharmacies, featuring prescription processing, patient records, inventory management, Canvas eye annotation, AI insights (Anthropic Claude), and Stripe subscription billing. Stack: React 18, Vite, Zustand, Node.js, Express, PostgreSQL.

Sri Lanka Railways Digital Platform (Softlogic / PelicanCube): National railways ticketing system including seat reservations, back-office management, and customer self-service portal. Stack: Next.js, NestJS, PostgreSQL.

Customer Success SaaS Platform – Core Product Areas (Velaris): Full-stack ownership of Success Plans, Tasks, Milestones, Files, Notes, and a drag-and-drop Scenario Workflow Builder. Stack: React.js, Node.js, PostgreSQL, AWS.

Australian School Analytics Dashboards (OctopusBI): Multi-tenant dashboards serving 200+ schools and 2,000+ users across Australia, integrated with Canvas LMS, Google Classroom, and Volkswagen. Stack: React.js, Next.js, Redux, Ant Design.

AI Glasses Try-On System (Final Year Project): Machine learning and deep learning model for virtual glasses recommendation and augmented try-on experience.

EDUCATION

Staffordshire University, United Kingdom

2020

BEng (Hons) Software Engineering – 2nd Class Honours

- Final Year Project: AI-powered Glasses Recommendation and Virtual Try-On System using Machine Learning and Deep Learning.

CERTIFICATIONS & AWARDS

- Udemy Certified – React.js, Hooks, React Router, and Redux (Full Developer Course)
- 1st Place – Remitbee Hackathon
- 2nd Runner-Up – APIIT Hackathon (Code Camp)